$$\begin{aligned}\left(x^2-\sqrt{1-x^2}\right)^5 &= {}^5\mathrm{C}_0\,(x^2)^5\left(\sqrt{1-x^2}\right)^0-{}^5\mathrm{C}_1\,(x^2)^4\left(\sqrt{1-x^2}\right)^1\\ &\quad+{}^5\mathrm{C}_2\,(x^2)^3\left(\sqrt{1-x^2}\right)^2-{}^5\mathrm{C}_3\,(x^2)^2\left(\sqrt{1-x^2}\right)^3\\ &\quad+{}^5\mathrm{C}_4\,(x^2)^1\left(\sqrt{1-x^2}\right)^4-{}^5\mathrm{C}_5\,(x^2)^0\left(\sqrt{1-x^2}\right)^5\\ &= x^{10}-5x^8\sqrt{1-x^2}+10x^6(1-x^2)-10x^4(1-x^2)\sqrt{1-x^2}\\ &\quad+5x^2(1-x^2)^2-(1-x^2)^2(\sqrt{1-x^2})\end{aligned}$$

Thus

$$\begin{aligned}\left(x^2+\sqrt{1-x^2}\right)^5+\left(x^2-\sqrt{1-x^2}\right)^5 &= 2[x^{10}+10x^6(1-x^2)+5x^2(1-x^2)^2]\\ &= 2[x^{10}+10x^6-10x^8+5x^2(1-2x^2+x^4)]\\ &= 2[x^{10}-10x^8+15x^6-10x^4+5x^2]\end{aligned}$$

Example 5.10 Using Binomial theorem, prove that 6^n-5n always leaves remainder 1 when divided by 25 for all positive integer n.

Solution:
To prove this it is enough to prove, $6^n-5n=25k+1$ for some integer k. We first consider the expansion

$$(1+x)^n={}^n\mathrm{C}_0+{}^n\mathrm{C}_1\,x+{}^n\mathrm{C}_2\,x^2+\cdots+{}^n\mathrm{C}_{n-1}\,x^{n-1}+{}^n\mathrm{C}_n\,x^n, n\in\mathbb{N}$$

Taking $x=5$ we get $(1+5)^n={}^n\mathrm{C}_0+{}^n\mathrm{C}_1\,5+{}^n\mathrm{C}_2\,5^2+\cdots+{}^n\mathrm{C}_{n-1}\,5^{n-1}+{}^n\mathrm{C}_n\,5^n$. The above equality reduces to $6^n=1+5n+25({}^n\mathrm{C}_2+5\,{}^n\mathrm{C}_3+\cdots+{}^n\mathrm{C}_n\,5^{n-2})$.
That is,

$$6^n-5n=1+25({}^n\mathrm{C}_2+5\,{}^n\mathrm{C}_3+\cdots+{}^n\mathrm{C}_n\,5^{n-2})=1+25k, k\in\mathbb{N}.$$

Thus 6^n-5n always leaves remainder 1 when divided by 25 for all positive integer n.

Example 5.11 Find the last two digits of the number 7^{400}.

Solution:
We have

$$\begin{aligned}7^{400} &= (7^2)^{200}=(50-1)^{200}\\ &= {}^{200}\mathrm{C}_0\,50^{200}-{}^{200}\mathrm{C}_1\,50^{199}+\cdots\\ &\quad+{}^{200}\mathrm{C}_{198}\,50^2(-1)^{198}+{}^{200}\mathrm{C}_{199}\,50(-1)^{199}+{}^{200}\mathrm{C}_{200}\,(-1)^{200}\\ &= 50^2\left({}^{200}\mathrm{C}_0\,50^{198}-{}^{200}\mathrm{C}_1\,50^{197}+\cdots+{}^{200}\mathrm{C}_{198}\,(-1)^{198}\right)-200\times50+1.\end{aligned}$$

As 50^2 and 200 are divisible by 100, the last two digits of 7^{400} is 0 1.